

Battery Divider & Regulator

The switched divider, plus the 3 V → 3V3 module that used to hang off it on wires, on one bank of a 5×30 solderable breadboard. The module's three pins land on 2.54 mm centres, so it drops into three rows of column D and the strips do the wiring that five flying leads were doing before.

10 rows • 1 bank

25 × 13 mm

10 components

1 jumper

7 external wires

module overhangs right

01 What changed from the divider-only board

Three rows go on at the top and everything below shifts down by three, so the eight divider rows are now rows 3–10. The regulator sits in column C of the new rows with **Vi in 3C**, **GND in 2C** and **Vo in 1C**, which leaves column D and column E of those rows for the leads that used to be soldered to the module's own pads.

Column D takes the pack: **pack + in 3D**, **pack – in 2D**. The negative lead is the thick one, so it gets a hole of its own rather than sharing with the ESP32's ground — and the two pack leads end up as adjacent holes in one column, which suits a twisted pair coming off the cell holder. **Read the warning in section 4 about that adjacency** before you solder either of them.

Two of the module's old wires disappear entirely. Its input no longer needs a lead from the pack *and* a lead to the divider, because **row 3 is already VBAT+** — dropping Vi into that row joins it to the pack wire, the solar wire, Q2's emitter and R6 through the strip. That is the whole gain: a node that existed anyway now does the joining.

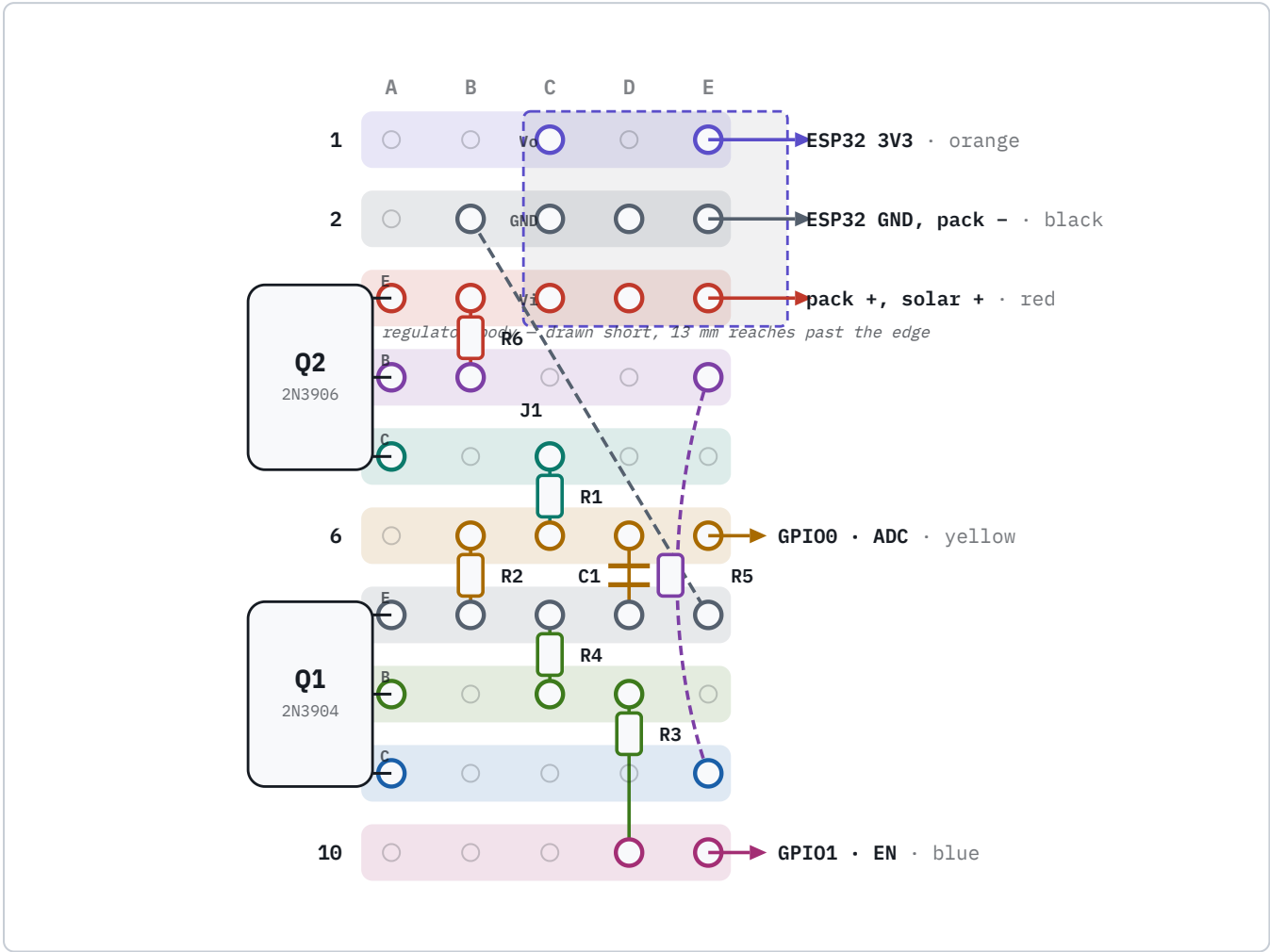
Its ground costs one jumper rather than none. The module's GND pin is in row 2 and the divider's ground is row 7, and strips are isolated, so **J1 bridges 2B → 7E**. That hole is free precisely because the ESP32's ground lead moved up to 2E.

Worth noticing what that leaves J1 carrying: **almost nothing**. The ESP32's supply loop runs regulator Vo → 1C → 1E → ESP32 → 2E → the row 2 strip → 2D pack –, and the regulator's own ground pin is on that same strip at 2C. So the whole of the load current stays inside row 2's copper, and the jumper carries only the divider's return — tens of microamps. A thin jumper is the right part for it, and that is a better arrangement than the single shared ground wire the earlier board used.

02 Layout

3V3 out GND · black VBAT+ · red Q2 base Q2 collector TAP · yellow

Q1 base Q1 collector EN · blue



Rows 1–10 of one bank, component side. The dashed outline is the regulator’s footprint, drawn short: its body really runs 13 mm from the pin row, so it overhangs the edge and lies over everything in columns D and E of rows 1–3 — five leads, the thick pack – among them. **J1** is the single jumper and **R5** the single flying resistor lead; both are dashed because they cross rows. Rows 2 and 7 share the ground colour because J1 makes them one net. Rows 11–30 and the second bank stay empty.

03 Nodes, row by row

Ten rows, nine nets — rows 2 and 7 are the same net once J1 is in.

ROW	NODE	HOLES
1	3V3 out	1C module Vo · 1E wire to ESP32 3V3
2	GND	2B J1 to 7E · 2C module GND · 2D wire to pack – (thick) · 2E wire to ESP32 GND

3	VBAT+	3A Q2 emitter · 3B R6 · 3C module Vi · 3D wire to pack + · 3E wire to solar +
4	Q2 base	4A Q2 base · 4B R6 · 4E R5
5	Q2 collector	5A Q2 collector · 5C R1
6	TAP → ADC	6B R2 · 6C R1 · 6D C1 · 6E wire to GPIO0
7	GND	7A Q1 emitter · 7B R2 · 7C R4 · 7D C1 · 7E J1 from row 2
8	Q1 base	8A Q1 base · 8C R4 · 8D R3
9	Q1 collector	9A Q1 collector · 9E R5
10	EN	10D R3 · 10E wire to GPIO1

04 Parts

REF	VALUE	HOLES	SPAN	MOUNTING
REG	3V3 module	1C / 2C / 3C	0.2"	Vo, GND, Vi — confirm which end is Vi first
J1	insulated wire	2B → 7E	0.5"	dress it round the edge, not across the middle
Q2	2N3906 PNP	3A / 4A / 5A	—	legs into rows 3, 4, 5 = E, B, C
Q1	2N3904 NPN	7A / 8A / 9A	—	legs into rows 7, 8, 9 = E, B, C
R1	330k	5C → 6C	0.1"	stand upright
R2	330k	6B → 7B	0.1"	stand upright
R3	10k	8D → 10D	0.2"	flat in 1/8W, upright in 1/4W — flies over 9D
R4	100k	7C → 8C	0.1"	stand upright
R5	100k	4E → 9E	0.5"	flat, sleeve the lead
R6	100k	3B → 4B	0.1"	stand upright
C1	10n ceramic	6D → 7D	0.1"	straight in — standard lead pitch

2D and 3D are the cell, one hole apart

Putting both pack leads in column D makes them tidy and makes them adjacent: **pack – in 2D and pack + in 3D, 2.54 mm apart with nothing between them.** A bridge there is a direct short across the cell — no resistor, no transistor, nothing to limit it — which on a lithium cell is the one fault on this board that can start a fire rather than just destroy a part.

So treat **2↔3** in the sweep as the test that matters most, do it before the pack ever

goes on, and look at those two joints under a loupe. Thick wire is harder to solder neatly in a 0.1" hole and more likely to leave a whisker, which is exactly the lead that sits there. If you would rather not have them adjacent at all, pack – also works in 2A – same row, same net, three holes further from pack +.

Which way round the module goes

The pin order reads left–right or right–left depending on which face is toward you. Fitted backwards, **Vo lands in row 3**, tying the regulator’s output straight to the pack, and the ESP32’s 3V3 lead ends up on the regulator’s input – pack voltage on a 3.3 V rail. The diagram shows the order, so this is only a reminder to mark the body before it goes in: **1C = Vo, 2C = GND, 3C = Vi**, output at the low-numbered end.

Pin order is the thing, not which way the body points

2N3904 and 2N3906 are both E–B–C: flat face toward you, legs down, left to right. Laid into rows 3→5 and 7→9, that means emitter in the lowest-numbered row. Confirm it with the diode test in section 6 before you solder anything else to those rows.

The header agrees now. It used to name Q1 as a 2N2222A; both `d_interface_batt.h` and `d_interface_cart.h` name the 3904, and the 2222 is gone from the codebase. The reason it had to go: in TO-92 the 2222 family is split, with ON Semi’s P2N2222A pinned E–B–C and the PN2222A pinned C–B–E, and the metal-can original is reversed again. That is a plausible cause of three dead hand-built boards. Pairing 3904 with 3906 leaves one footprint to get right instead of two.

05 Solder order

The divider goes in exactly as before. Then the wires, then the two pack leads, then the module **last** – once rows 1–3 have been proved open, because the body covers everything in columns D and E of those rows and you will not get a probe in afterwards.

- 1 Cut the board to 12 rows and keep one bank. Fit **Q2** into 3A/4A/5A and **R6** upright between 3B and 4B.

Check – **diode-test Q2**, and **3↔4 ≈ 100k**.

- 2 Fit **Q1** into 7A/8A/9A and **R4** upright between 7C and 8C.

Check – **diode-test Q1**, and **7↔8 ≈ 100k**.

- 3 Fit **R1** upright (5C–6C), **R2** upright (6B–7B), and **C1** straight into 6D–7D.
Check — **5↔6 = 330k**, **6↔7 = 330k**.
- 4 Fit **R3** (8D→10D), clearing hole 9D.
Check — **8↔10 = 10k**, and **9↔10 open**.
- 5 Fit **R5** (4E→9E) with sleeving over the long lead, then **J1** from 2B to 7E.
Check — **4↔9 = 100k**, **2↔7 = 0Ω**, and the **full adjacent-row sweep** including **3↔7 open**.
- 6 Fit the five signal and supply wires first: 6E yellow, 10E blue, 1E orange, 2E black, 3E solar + red. Dress everything in rows 1–3 flat and outward — all of column D and E there ends up under the module.
Check — **1↔2**, **1↔3** and **2↔3** all open, with nothing in column C yet.
- 7 Fit the two pack leads last of the wires, **pack – in 2D and pack + in 3D**, and inspect both joints under a loupe before anything else.
Check — **2↔3 open**. This is the cell short. Do not connect a cell until it reads open.
- 8 Mark the module’s output end, then fit it into 3C / 2C / 1C.
Check — **1↔3 open** (this is the one that would kill the ESP32), **1↔2** reads the regulator’s own output impedance and not a short, and **3↔7** and **2↔3** still open.

06 Bench tests

Resistance, unpowered

Probe any free hole in each row — all five holes in a row are the same node. Every adjacent pair is listed, because with no empty rows this sweep is also the solder-bridge test: a bridge reads near zero where the table says otherwise.

ROWS	EXPECT	NOTES
1 ↔ 3	open	The new one that matters. 3V3 out against the pack. A short here — a backwards module, or a bridge across rows 1–3 in column D — puts cell voltage on the ESP32’s 3V3 rail.
1 ↔ 2	not a short	the regulator’s own output impedance, often a few kΩ falling as its output cap charges — what matters is that it is not 0Ω
2 ↔ 3	not a short	The cell short. Pack – in 2D sits one hole from pack + in 3D, so a bridge here is the battery onto itself. Electrically you are reading the regulator input to ground,

which should be a high reading or a junction — never 0Ω . Check it before the pack goes on and again after the module.

2 ↔ 7	0Ω	J1 — the one connection on this board that is <i>meant</i> to be a dead short. It carries only the divider's return, so thin wire is correct
3 ↔ 4	$\approx 100k$	R6, in parallel with Q2's base-emitter junction — one probe polarity reads lower, both are fine
4 ↔ 5	a junction	Q2 base-collector only, no resistor — expect a diode, not a resistance
5 ↔ 6	330k	R1
6 ↔ 7	330k	R2 — the reading creeps upward as C1 charges
7 ↔ 8	$\approx 100k$	R4, in parallel with Q1's base-emitter junction
8 ↔ 9	a junction	Q1 base-collector only, no resistor
9 ↔ 10	open	nothing joins Q1's collector to the enable line
4 ↔ 9	100k	R5
8 ↔ 10	10k	R3
3 ↔ 7	open, $>1M$	Still the one that can hurt a cell. Pack against ground. With J1 in, row 2 is ground too, which is why 2↔3 above is the same fault measured from the other end.

Transistor orientation, diode mode, in circuit

The 100k parts around each junction are far higher than the junction itself, so the readings come through clearly without unsoldering anything.

PART	BASE IS	EXPECT $\sim 0.7V$	EXPECT OPEN
Q1 NPN	common anode	red 8 / black 7 red 8 / black 9	both reversed
Q2 PNP	common cathode	black 4 / red 3 black 4 / red 5	both reversed

Powered

Use a bench supply at 4.00V with the limit set to 10mA, not a cell, and **nothing connected to the ESP32 leads yet**. If something is wired wrong the supply folds back instead of a lithium cell delivering into it. Feed it at 3D and ground at 2D — the pack holes, so the supply sees exactly what a cell would.

STATE	EXPECT	NOTES
power on, EN open	row 1 reads $3.3V \pm 0.1$	do this one first — if row 1 reads 4V the module is backwards, and 10mA of limit is what saves the ESP32 you have not connected yet

ESP32 you have not connected yet

module quiescent draw	tens of μA	whatever its datasheet says — worth measuring once, since with the ESP32 asleep this and the module's own losses are most of what the cell sees
EN open (or at 0V)	row 9 reads 0V	divider off — measure against the module's own draw, which no longer makes the total unmeasurable
EN at 3.3V	row 9 reads just under 2.00V	~45 μA from the supply, plus ~260 μA sourced by the GPIO through R3
EN back to open	row 9 under 100mV within 25ms	this is the slow path — R2 alone draining C1, which is what BATTERY_SETTLE_MS exists to cover

Row 9 reads a little *under* half the supply, and the shortfall is Q2's saturation drop — which is the thing BATTERY_OFFSET_MV exists to trim out, so measure it rather than chase it. At 45 μA for a 25ms settle, one reading a minute costs about 0.02 μAh an hour, which against an IMR14500's 500mAh is the point of the whole circuit. The regulator's own quiescent current sits beside it as the larger number, which is worth measuring once so you know the floor the sleeping node starts from.

07 Then let the driver accept it

The pass criteria are already written down in your own probe constants, so the board does not need a judgement call:

CRITERION	CONSTANT	LIMIT
Divider off, reading at the pin	BATTERY_PROBE_OFF_MAX_MV	≤ 150
On minus off, so the pin follows the enable	BATTERY_PROBE_DELTA_MIN_MV	≥ 300
Two off readings agree	BATTERY_PROBE_OFF_DIFF_MV	≤ 100
Pack voltage is plausible	BATTERY_PROBE_MV_MIN / MAX	2000–4500

A good board reports neither BATTERY_STATUS_NO_DIVIDER nor BATTERY_STATUS_DIVIDER_FAULT. Once it passes, read a known cell voltage with a meter and set BATTERY_OFFSET_MV for this board — your carrier sits at 155.

08 Notes

- **The module overhangs, and from column C it covers more than it did from D.**
Its body runs 13 mm from the pin row, five hole-pitches, so facing outward it lies over *both* column D and column E of rows 1–3 and still reaches about 8 mm past the board edge. Turned inward instead it goes back over columns B and A,

where it would foul R6 standing upright in 3B — so outward is the only option.

- **Five leads now sit under it**, including the thick pack –. Soldering is unaffected, since every joint is on the track side, but a thick lead dressed on the component side will hold the module off the board. Two ways out: bring pack + and pack – in from the *track* side so they leave downward, or fit the module on the track side as well, so its body hangs under the board and the whole component face stays clear. You said it mounts either way up, and that second option is the tidier one if the housing has depth below rather than beside.
- **J1 should go round the outside, not across the face.** The diagram draws it straight from 2B to 7E to show the topology, but a wire taking that path passes directly over the upright R1, R2 and C1 and over row 6, the 165k ADC node. Route it past the column E edge instead. It is insulated, so proximity is harmless — contact and strain are not.
- **Buy 1/8W resistors for this version.** Five parts stand on end, and a 1/4W body puts them about 9mm above the board. A 1/8W body is 3.2mm, which stands at roughly 5mm, level with the TO-92 cans, and lets R3 lie flat as well. Nothing here dissipates anything: R3 is the worst at 0.7mW.
- **Rows 3 and 7 use all five holes**, and row 3 is the busiest node on the board: pack, solar, regulator input, Q2's emitter and R6. Row 2 has one spare hole at 2A — the only slack left in the top three rows, and the alternative home for pack – if you want it away from pack +. If you want slack on ground generally, one jumper from row 7 across the centre channel gives five more GND holes in the second bank.
- **Your error margin is spent.** Every row is live, so a bridge between any neighbouring pair is a real fault. Four are worth more than a rework: **1–2** and **1–3** around the regulator, **5–6** shorts R1 and puts the full pack voltage on GPIO0, and **4–5** leaves the divider permanently energised, draining the cell around the clock. The sweep in section 6 catches all four — treat it as mandatory, and inspect between the rows under a loupe.
- **The 3V3 rail is unswitched, and that is the design.** The ESP32 is what goes to sleep, so the rail feeding it has no business being gated — there would be nothing left to wake it. The divider is switched for a different reason: it hangs off VBAT directly, upstream of the regulator and permanently live, so without Q1 and Q2 its 330k pair would sit across the cell around the clock. One is gated because the load sleeps; the other because nothing else would ever turn it off.
- **Keep the ADC wire short, and not bundled with the EN wire.** Row 6 is a 165k node, high enough to pick up whatever is next to it. C1 is what makes it measurable at all.